

# COMP2005

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- Exam Revision

# The Quiz (30%)

- 1 hour Quiz  
in the lab - Thursday 30<sup>th</sup> April ( **Time and Place TBC**)
- Answer ALL questions
- Exam Format : ExamSys (closed book)
- Focus on image processing methods
  - What they do
  - How they work
  - When they are appropriate
  - *Knowledge, Comprehension, Application*
  - Questions are (loosely) structured, topic area indicated

**Read and answer the question**  
**Take no. of marks available into account**

# What we covered in this module

- Image formation and acquisition
  - Background material: cameras, Bayer patterns
  - Basic terminology: sampling and quantisation
  - A little image processing: re-sampling, re-sizing, re-quantisation
- Colour spaces
  - RGB, HSV, LAB etc
  - Image pre-processing: choosing a colour space is a key step in practical applications, but not really IP
- Histogram methods
  - Histogram equalisation
- Point transforms
  - Gain, bias, contrast stretching, gamma correction
  - Simplest methods, *but useful*

# The End

- Spatial Filtering
  - Convolution is key
  - Noise removal: mean, Gaussian filtering
  - Enhancement: unsharp masking, Laplacian filtering
  - Edge detection: Roberts, Sobel, Canny

- Non-linear filtering
  - Median, anisotropic diffusion, bilateral filtering

*Linear and non-linear filters are at the heart of the image processing toolbox*

- Thresholding and Binary Images
  - Otsu, Unimodal thresholding
  - Connected components, morphology (erosion, dilation, opening and closing)

# The End

- Segmentation
  - Region growing
  - Split and Merge (quadtrees)
  - Watersheds

A set of tools that can be used to create image processing pipelines

# Module Aims

- To introduce the fundamentals of digital image processing - theory and practice.
  - **Assessed by exam: how do techniques work, and what do they do**
- To gain practical experience in writing programs to manipulate digital images.
  - **Assessed by coursework – no coding in the exam**
- To lay the foundation for studying advanced topics in related fields

# Sample Questions

## Question 1:

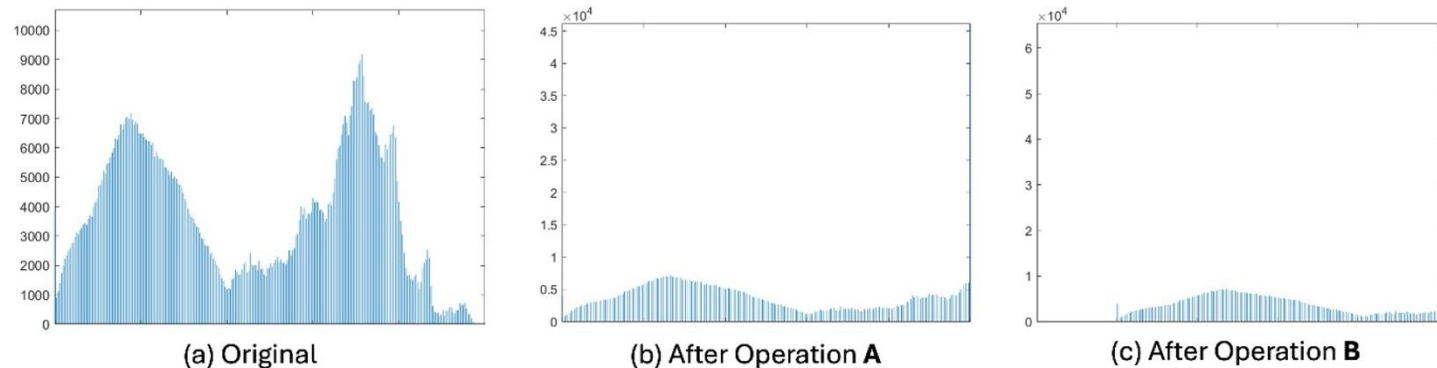


Figure 1

a. This question has two parts (P1, P2)

**P1.** Figure 1 show the process of a linear transformation. Given the original

histogram in (a), what process is happening in (b)  and (c)  respectively?

**P2.** Which of the following statements **accurately** define (b) and (c), respectively:

- i. (b) controls contrast while (c) controls brightness
- ii. (b) controls hue while (c) controls value
- iii. (b) controls brightness while (c) controls contrast
- iv. (b) controls saturation while (c) controls hue
- v. (b) controls intensity while (c) controls pixelization |

## Sample Questions

### Question 2:



Figure 2

Figure 2 shows an image experiencing salt & pepper noise. Based on the selections given

in the drop-menu below, which filter would you recommend for noise reduction and why?

- i. Mean filtering because it can make the resulting image noise free and sharp.
- ii. Gaussian filtering because it can enhance the properties of the tulip in the resulting image.
- iii. Anisotropic diffusion because it reduces noise and preserve the tulip's edges in the resulting image.
- iv. Median filtering because its preserves the real pixel values in the resulting image.



## Sample Questions

### Question 3:

A 3-bit per pixel image has a normalised histogram as listed in Table 1. Apply histogram equalisation to this data and complete the remaining columns which will map from the input pixel values to output pixel

**Table 1**

| Input Pixel Value | Normalised Frequency | CDF  | Output Pixel Value (rounded) | Output Normalised Frequency |
|-------------------|----------------------|------|------------------------------|-----------------------------|
| 0                 | 0.05                 | 0.05 |                              |                             |
| 1                 | 0.1                  | 0.15 |                              |                             |
| 2                 | 0                    | 0.15 |                              |                             |
| 3                 | 0.2                  | 0.35 |                              |                             |
| 4                 | 0.2                  | 0.55 |                              |                             |
| 5                 | 0.2                  | 0.75 |                              |                             |
| 6                 | 0.05                 | 0.8  |                              |                             |
| 7                 | 0.05                 | 0.85 |                              |                             |
| 8                 | 0.05                 | 0.9  |                              |                             |
| 9                 | 0.1                  | 1.0  |                              |                             |

# Good Luck

**Please check the rooms and give yourself half an hour  
early arrival to the exam**